MIL-C-18759B (AS)

19 June 1968

Superseding
MIL-C-18759A (AS)
6 June 1967

MILITARY SPECIFICATION
CARTRIDGE CASE, STEEL, 20-MILLIMETER
MARK 5 AND MODS
(FOR 20MM A/C GUNS)

1. SCOPE

1.1 This specification covers the manufacture and preparation for delivery of the Cartridge Case, Steel, 20-Millimeter Mark 5 and Mods and the methods of examination and tests upon which product acceptance shall be based.

2. APPLICABLE DOCUMENTS

2.1 Government Documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

Federal

QQ-Z-325	Zinc Plating (Electrodesposited)
PPP-B-636	Boxes Fiberboard

Military

MIL-S-3289	Steel, Plate and Disk Carbon, Forging Quality
MIL-Q-9858	Quality Program Requirements
MIL-I-45607	Inspection Equipment, Supply and Maintenance of
MIL-C-45662	Calibration System Requirements

STANDARDS

Federal

FED-STD-151	Metals, Test Methods

FSC8140

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

Military

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage

DRAWINGS

Naval Ordnance Systems Command

NAVAIRSYSCOM 517534	20MM Cartridge Case, MK 5 MOD 0 Steel
LD 289036	20MM Cartridge Case, Steel, MK 5 MOD 0

(And LD's, drawings, specifications, standards and publications listed thereon.)

Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply:

AMERICAN SOCIETY FOR TESTING AND MATERIALS

ASTM E3-62	Preparation of Metallographic Specimens

Address request for copies to American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pa. 19103

AMERICAN SOCIETY OF MECHANICAL ENGINEERS

ASA B46.1-1962	SURFACE TEXTURE (Surface Roughness, Waviness & Lay)

Address request for copies to American Society of Mechanical Engineers, United Engineering Center, 345 East 47th Street, New York, N.Y.

Technical Society and Technical Association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.

3. REQUIREMENTS

3.1 Inspection and Manufacture. The cartridge cases shall be manufactured, examined, tested and packed in accordance with contract documents, NAVAIRSYSCOM Drawing 517534, this specification, and all LD's, drawings, specifications, standards and publications listed herein.

3.1.1 Preproduction sample. An acceptable preproduction sample of cartridge

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

cases shall be furnished by the supplier (see 4.1.1). The cartridges and all materials shall be of single type, grade, class, size and composition and shall comply with all applicable provisions of this document. The preproduction sample shall not be applied as part of the quantity specified for delivery by the contract or order. Production of the item prior to acceptance of the preproduction sample and authorization by procuring agency shall be at the supplier's risk.

3.1.2 General

3.1.2.1 Material. Materials used in the manufacture of the cartridge case shall be in accordance with Specification MIL-S-3289, except the supplier may use steel of chemical composition differing from the ranges specified in Specification MIL-S-3289, initially or during production, provided that such a change has been proven by the successful performance of a preproduction sample and the change is approved by the procuring activity. A chemical analysis report of the material representing each preproduction sample and each production lot shall be submitted by the supplier to the procuring activity.

3.1.2.2 Machining. Machining of cartridge cases shall not result in objectionable burrs, gouges, or other imperfections which will affect their serviceability.

3.1.2.3 Surfaces. All cartridge case surfaces shall be free from injurious defects such as blisters, cracks, slivers, seams, cold shuts, folds and other imperfections and shall be smooth in appearance after protective coating. "Injurious" shall be defined as exceeding 10% of the wall thickness, and shall apply to any and all imperfections.

3.1.2.4 Protective coating. The coating process shall conform to Specification QQ-Z-325, Type II, Class 3, except that the coating shall have a minimum thickness of 0.0002-inch on the interior surfaces, 0.0003-inch on the exterior body surface, and a maximum thickness of 0.0008-inch on either surface.

3.2 Examinations. Examinations shall be conducted to ensure cartridge cases meet all requirements of Drawing Number 517534.

3.3 Performance and product characteristics

3.3.1 Performance

3.3.1.1 Ballistic. The cartridge case shall meet the following requirements:

a. Gun Function in both the MK-11 and MK-12 machine guns: Load and Eject

b. No longitudinal or circumferential splits, cracks, or fractures other than flange cracks as specified below, before and after ballistic tests.

c. The flange area in Regions "A" and "AB" shall not shear during gun chamber loading, extraction, and ejection. (Flange cracks other than complete penetration of case wall are to be reported for information only and are not cause for rejection.) Regions "A" and "AB" are defined in Drawing 517534.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

3.3.2 Product.

3.3.2.1 Microstructure. All regions of the cartridge case shall display a uniform fine grain microstructure compatible with the type of conditioning used without evidence of surface carburization or complete decarburization which would, in any way, adversely affect gun functioning or reduce the rate of fire. Decarb shall not exceed .010 inches complete and/or .020 inches total affected area except in instances where adverse effect on gun function is suspected under which circumstances partial decarb shall be restricted to .003 inches (depth) maximum.

3.3.2.2 Hardness. Hardness values for test regions of the cartridge case shall be in accordance with Drawing 517534. Region A of the cartridge case head shall be annealed by an appropriate method to reduce the hardness in the head area to a maximum value of 79 Rockwell (30-T). The area affected by this annealing treatment must not extend beyond the limits specified for Region A.

3.3.2.3 Tension Test. The yield strength of the cartridge case of Region B, as shown on Drawing 517534, should be 135,000 pounds per square inch (min.). The elongation of Region B shall be 6.0 percent in 1-inch (min.).

3.3.2.4 Corrosion. The cartridge case shall meet the requirements of this specification. (See 4.3.2.4)

3.4 Workmanship. The product, in all stages of production, shall be processed under such inspection control as to assure uniformity in quality. All components shall be free of chips, dirt, grit, or other foreign material. The cleaning methods used shall not be injurious to any of the parts nor shall the parts be contaminated by the cleaning agents employed. The presence of scratches which extend through the protective coating shall render a case liable to rejection.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection and requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Preproduction lot. The supplier shall deliver a preproduction lot of 1000 cartridge cases to the destination specified (Section 6.2C) for examination, testing, evaluation and acceptance per Table I and 4.2 before general production on the contract is approved. These cartridge cases shall be made by the same processes the supplier will use in the fullfillment of the contract. A new preproduction lot shall be submitted for approval before any major changes in the manufacturing process will be accepted by the procuring agency.

4.1.2 Production Lot. A production lot shall consist of all cartridge cases from one line or operation offered for delivery at one time and shall not exceed 50,000 units. The test sample shall be drawn from each production lot. Production from various lines or operations shall not be combined into production lots unless statistical control charts maintained during the production period for each of

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

the various lines or operations are submitted with combined production lots and homogeneity is demonstrated on the charts. Production lots shall be examined as required in 4.2 and tested and accepted in conformance with Table I. The characteristics verified by the tests and examination herein are classified as Critical, Major or Minor, in accordance with Drawing 517534. Tests and examinations that verify Critical characteristics are identified by the symbol (C), and Major characteristics by the symbol (M). Test and examinations which are not annotated with a classification code are classified Minor.

TABLE I

INSPECTION PLANS

REFERENCE PARAGRAPH	TESTS	PREPRODUCTION SAMPLE		PRODUCTION LOT	
(Blank)	(Blank)	Sample Size	Acceptance Criteria	Sample Size	Acceptance Criteria
4.3.1.1 3.3.1.1	Ballistic Tests	1000 Cartridge Cases	a. Ac 0 Re 1 b. Ac 0 Re 1 c. Ac 0 Re 1	MIL-STD-105 Inspection Level II	a. Ac 0 Re 1 b. Ac 0 Re 1 c. Ac 0 Re 1
4.3.2.1 3.3.2.1 4.3.2.2 3.3.2.2	Microstructure Test and Hardness Test	13 Cartridge Cases	Ac 0 Re 1	MIL-STD-105 Inspection Level S-2	AQL 1.0
4.3.2.3 3.3.2.3	Tension Test	13 Cartridge Cases	Ac 0 Re 1	MIL-STD-105 Inspection Level S-2	AQL 1.0
4.3.2.4 3.3.2.4	Corrosion Test	13 Cartridge Cases	Ac 0 Re 1	MIL-STD-105 Inspection Level S-3	AQL 1.0

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

TABLE II

Type of Sampling	Number of Rounds	Charge Pressure (Propellant)	Temperature of Rounds (F^{o})
Preproduction	300	Service	-65
	300	Service	+70
	300	Service	+165
	25	Proof	+70
	75	Misc. Tests	As required
Production	10	Proof	+70
	1/3 of Remainder of Sample	Service	-65
	1/3 of Remainder of Sample	Service	+70
	1/3 of Remainder of Sample	Service	+165

Each round for testing at service pressure shall be fired under automatic rapid fire, and each round for testing at proof pressure shall be fired in single shot, unless otherwise directed by the procuring activity. All rounds shall be fired with charges and at temperatures as shown in Table II. The tests shall be performed with a weapon of service quality. The gun mechanism, barrel, and feeder mechanism shall be of current service design and in an operating condition satisfactory for unrestricted use. All components of the gun mechanism shall have been prepared for action in accordance with pertinent ordnance publications. A weapon of service quality shall be used for all ballistic testing of cartridge cases at service pressure. For testing at proof pressure an accuracy or Mann barrel and associated equipment shall be used. All cartridge cases tested under automatic rapid fire action shall be extracted and ejected, and in no way cause the weapon to cease firing. When fired in single shot action, the cartridge cases with proof charges shall be capable of being extracted by normal methods. Defectives are cartridge cases failing to meet the requirements of 3.3.1.1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

4.1.3 Test equipment and gages. All test equipment and gages shall be supplied and maintained by the supplier as specified in Specification MIL-I-45607. The supplier shall maintain a calibration system as specified in Specification MIL-C-45662, unless otherwise specified in the contract. However, a supplier currently under the requirements of MIL-Q-9858 shall be governed by the requirements of MIL-Q-9858 for items procured in accordance with this specification.

4.2 Examinations. Sampling and inspection characteristics, unless otherwise noted, shall be in accordance with MIL-STD-105 (Table IIa); sampling for characteristics as specified on Drawing 517534 shall be as follows:

(a) For critical defects a single sampling is prescribed (double or multiple sampling and reduced inspection are prohibited). Sample size shall be as follows:

(1) For inspection of lot size of 125 or less, use 100 percent of the lot.

(2) For inspection of lot sizes of 126 to 3,200, use 125 units.

(3) For inspection of lot sizes greater than 3,200, use the sample size from Table II of MIL-STD-105.

(b) Reject the lot for one or more CRITICAL defects.

(c) The AQL for MAJOR defects shall be 1.0 percent by class and for MINOR defects 2.5 percent by class unless otherwise specified.

4.2.1 Cartridge case examination. Examination shall be made to ascertain that the cartridge cases meet the requirements of Drawing Number 517534. Only cartridge cases containing no defectives shall be used for further testing or returned to the lot.

4.2.2 Packing and marking examination. The packing of the cartridge case and marking of the pack shall conform to Section 5 of this specification.

4.3 Tests. Random samples of cartridge cases shall be selected and tested as specified in Table I as applicable.

4.3.1 Performance test.

4.3.1.1 Ballistic testing. The cartridge cases shall be loaded and crimped as for service use. For preconditioning and firing of samples refer to Table II.

4.3.1.1.1 Miscellaneous tests. The testing activity may conduct additional tests as required to assist component manufacturers and assembly activities to maintain quality control. Such tests shall be authorized by the procuring agency. Test results shall not be used as acceptance or rejection criteria, but shall be included in Ballistic test reports for information purposes.

4.3.2 Product tests.

4.3.2.1 Microstructure test. The cartridge case shall be prepared for microstructure examination in accordance with ASTM E3-62; paragraphs 2 through 6, inclusive, 27 and 29. The presence of surface carburization and complete decarburization and extent of partial decarburization shall be determined by

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

microscopic examination of the cartridge cases at a magnification of 100 diameters minimum. Photomicrographs with the magnification ratio permanently recorded shall be prepared for each cartridge case tested. Defectives are cartridge cases failing to meet the requirements of 3.3.2.1.

4.3.2.2 Hardness test. A sufficient number of hardness determinations shall be made on two quadrants of each cartridge case so as to represent adequately the hardness of the regions shown on Drawing 517534. Requirements for any region apply to the metal throughout the region as indicated on Drawing 517534. Cases must be sectioned for hardness tests when the values obtained on the internal or external surfaces do not indicate the true hardness of the region. When a specimen is required it shall be sectioned longitudinally from the specified region and properly prepared on both sides, free from dirt, scale, and surface irregularities. Standard methods for the preparation of surface for hardness determination shall be employed. Any method of hardness determination that gives values which can be accurately converted to the specified Rockwell scale shall be used. (See Fed. Test Method Std. No. 151A - Method 241.2) Defectives are cartridge cases failing to meet the requirement of 3.3.2.2.

4.3.2.3 Tension test. Tension tests shall be required on the cartridge cases, unless otherwise specified by the procuring activity. Yield strength and elongation tests of the cartridge cases shall be made on circumferential specimens cut from Region B of the case as shown on Drawing 517534. Tension test specimens shall be straightened by pressing and rolling and then milled to the standard test size for sheet metal, or a size as may be specified in the detail specification. They may be stress relieved at a temperature of 100°F or more, below that received in the tempering of the cartridge case. With the approval of the Government the test may be made on a tension test specimen in which the thickness represents the original cross-section of the case in the specified region. The tension tests shall be performed on a testing machine, together with the necessary accessories for accurately determining yield strength and elongation, which has been approved by the Government. The rate of stress applicable of the load in determining yield strength shall not exceed 50,000 p.s.i. per minute. (See Fed Test Method Std No. 151A - Method 211.1) Defectives are cartridge cases failing to meet the requirements of 3.3.2.3.

4.3.2.4 Corrosion test. Corrosion tests shall be conducted on cartridge cases (plated parts) as specified in Federal Specification QQ-Z-325; salt spray requirements shall be as specified except that salt spray exposure shall be for 120 hours at ambient temperature using a 20 percent salt solution. Defectives are cartridge cases failing to meet the requirements of QQ-Z-325.

5. PREPARATION FOR DELIVERY

5.1 Packing

5.1.1 Level A - Not Applicable

5.1.2 Level C - The cartridge cases shall be packed by the contractor for domestic shipment in corrugated fiber boxes, in conforming to Specification PPP-B-636. Packing shall protect the primer hole, its counterbore, and the mouth of the case, and to separate the cases in a suitable manner so that they will not be damaged in shipment.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

5.2 Marking

5.2.1 Normal marking. In addition to any requirements of the contract or order, containers shall be marked in accordance with Standard MIL-STD-129.

6. NOTES

6.1 Intended use. The Cartridge Case, Steel, 20-Millimeter Mark 5 and Mods is for use with primers, propellant charges, and loaded projectiles to form complete rounds for use in 20-Millimeter aircraft guns.

6.2 Ordering data. Procurement documents shall specif.

a. Title, type and symbol and date of this specification.

b. Production lot size.

c. Test facility where preproduction samples are to be shipped for evaluation.

6.3 Definitions. The meaning of the terms used in this specification shall be listed as below.

6.3.1 The term "Cartridge Case" shall mean Cartridge Case, Steel, 20-Millimeter Mark 5 and Mods as depicted on Drawing 517534.

6.4.2 The term "Ejected" means that the cartridge case is thrown free of the breech to permit loading of the subsequent round of ammunition in a manner that in no way interferes with continuous action of the weapon.

6.4 Conflicting requirements. Conflicting requirements arising between this specification and any specification, publication or drawing listed herein shall be referred in writing to the procuring agency for clarification.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-C-18759(AS) Cartridge Case, Steel, 20 Millimeter Mk 5 & Mods (For 20MM A/C Guns)

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426

C-8878

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.

FOLD

DEPARTMENT OF THE NAVY
Naval Air Systems Command
Washington, D.C. 20360

POSTAGE AND FEES PAID
NAVY DEPARTMENT

OFFICIAL BUSINESS

NAVAL AIR SYSTEMS COMMAND (AIR-52021)
DEPARTMENT OF THE NAVY
WASHINGTON, D. C. 20360

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-18T11:55Z
Check the source to verify that this is the current version before use.